Assignment 2 Answers and Explanation

Task2

Suitable IP range for HQ and branch

HQ supports up to 8000 devices and branch supports up to 500 devices.

Private block of 10.10.0.0/16 (IPv4 address)

An IPv4 has 32 bits, so to find host bits:

32 bits - 16 network bits = 16 host bits

To calculate how many usable IP addresses a subnet provides:

Usable hosts = 2^h - 2

2^13 - 2 = 8190 (supports more than 8000 devices for HQ)

2^9 - 2 = 510 (supports more than 500 devices for branch)

To find network bits, subtract from 32 bits:

32 bits - 13 host bits = /19 (HQ)

32 bits - 9 host bits = /23 (branch)

IPv4 has 4 octets (32 bits) so each octet = 8 bits:

/19 takes up 2 octets + 3 bits of 3rd octet = 224

/23 takes up 2 octets + 7 bits of 3rd octet = 254

Hence subnet masks are:

/19 = 255.255.224.0

/23 = 255.255.254.0

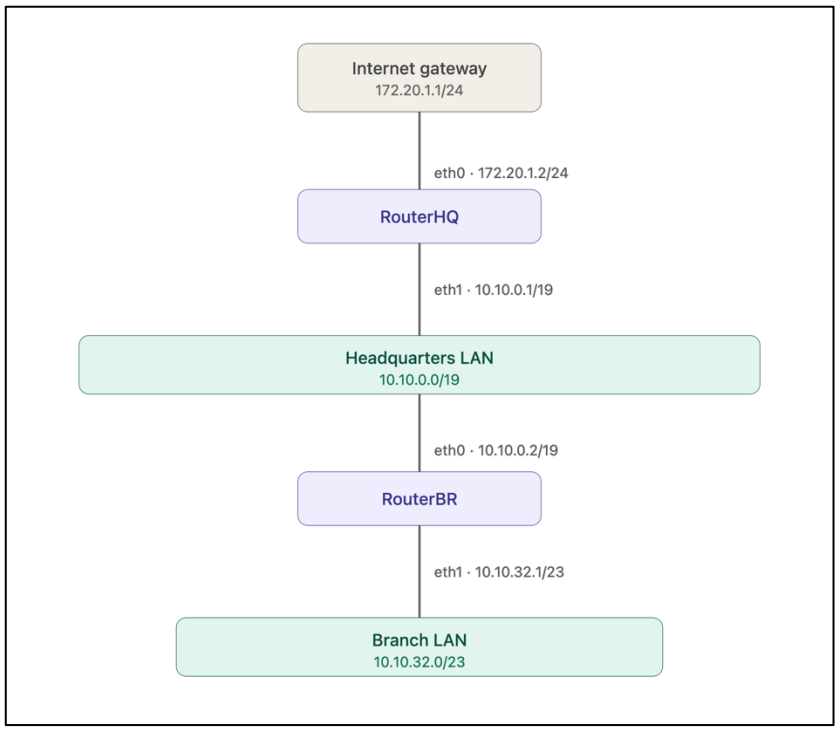
To calculate block size:

256 - 224 = 32

256 - 254 = 2

	HQ Range	Branch Range
Network	10.10.0.0	10.10.32.0
Usable Range	10.10.0.1 - 10.10.31.254	10.10.32.1 - 10.10.33.254
Broadcast	10.10.31.255	10.10.33.255

Network Diagram



RouterHQ Routing Table

Destination	Gateway	Genmask	Flags	Iface
10.10.0.0	*	255.255.224.0	U	eth1
10.10.32.0	10.10.0.2	255.255.254.0	UG	eth1
172.20.1.0	*	255.255.255.0	U	eth0
default	172.20.1.1	0.0.0.0	UG	eth0